

4 Parametric Learning for System Identification and Imitation Learning

Consider a time-series

$$Y_t = h^*(X_t) + E_t, \quad t = 1, \dots, T. \quad (4.1)$$

with **outputs** (or **targets**) Y_t , **inputs** (or **covariates**) X_t , and **noise** E_t . The function h^* is the unknown **regression function** that we want to learn from data.

We will make the following assumptions in this lecture:

Assumption 4.1 (Parametric Regression Function). We assume that the regression function h^* is a **known function** h of an unknown parameter θ , i.e., that $h^*(\cdot) = h(\cdot; \theta^*)$.

Under Assumption 4.1, the learning problem reduces to identifying the unknown parameter θ^* . This motivates the introduction of the noise process $\{E_t\}$ to the time-series model (4.1), as without noise, θ^* can be recovered **exactly** from finitely many samples (under suitable excitation assumptions).

We make the following mild assumption on the statistics of the noise process $\{E_t\}$.

Assumption 4.2 (MDS Process Noise). The noise process $\{E_t\}$ is a **martingale difference sequence** (MDS) relative to the filtration $\mathcal{F}_t := \sigma(X_0, \dots, X_t, E_0, \dots, E_{t-1})$, i.e., the information available at time t . In particular, this means that

$$\mathbb{E}[E_t \mid \mathcal{F}_t] = 0. \quad (4.2)$$

We further assume that the noise process has conditional covariance given by

$$\mathbb{E}[E_t E_t^\top \mid \mathcal{F}_t] = \Sigma \succ 0. \quad (4.3)$$

Remark 4.3. Importantly, $\{E_t\}$ need **not** be i.i.d., Gaussian, or even stationary!

This model captures our two motivating problems, namely system identification and imitation learning, through suitable definitions of Y , X , h^* , and E .

Example 4.4 (System Identification as Time-Series Prediction). Consider a nonlinear dynamical system $X_{t+1} = f^*(X_t) + W_t$, where X is the state, f^* the dynamics map, and W_t the process noise. This can be put in the form (4.1) by setting $Y_t = X_{t+1}$, $h^* = f^*$, and $E_t = W_t$.

Example 4.5 (Imitation Learning as Time-Series Prediction). Consider an expert policy $\pi^*(X)$ deployed in closed loop on a dynamical system $X_+ = f(X, U)$ by setting

$U = \pi^*(X) + N$, where N is an unknown input noise. Denote the resulting expert trajectory by X_t^* , and the state/action pairs by $\{X_t^*, U_t^*\}_t$, resulting in the input/output relationship $U_t^* = \pi^*(X_t^*) + N_t$. This matches the form (4.1) with $Y_t = U_t^*$, $h^* = \pi^*$, and $E_t = N_t$.

The rest of this chapter is devoted to the analysis of the system identification example. However, all results translate naturally to the imitation learning and general time-series prediction setting, possibly with slightly adjusted assumptions.

4.1 System Identification as Time-Series Prediction

Consider an unknown nonlinear dynamical system

$$X_{t+1} = f(X_t; \theta^*) + W_t \quad (4.4)$$

where $W_t \in \mathbb{R}^n$ is a stochastic process noise satisfying Assumption 4.2, f is a known function, and θ^* is an unknown parameter to be identified from trajectory data $\{X_0, X_1, \dots, X_T\}$.

We consider three examples of the dynamics (4.4), in increasing order of generality:

$$(a) \quad X_+ = \theta^* X + W, \quad \theta^* \text{ unknown} \quad (\text{LTI system}), \quad (4.5)$$

$$(b) \quad X_+ = \theta^* \Phi(X) + W, \quad \theta^* \text{ unknown, } \Phi \text{ known} \quad (\text{linear in features}), \quad (4.6)$$

$$(c) \quad X_+ = f(X; \theta^*) + W, \quad \theta^* \text{ unknown, } f \text{ known} \quad (\text{general nonlinear}). \quad (4.7)$$

4.2 The Prediction Error Estimator

We will study the behavior of the **prediction error least squares estimator**.

Definition 4.6 (Prediction error least squares estimator). Given trajectory data $\{X_0, \dots, X_T\}$, the **prediction error least squares (PELS)** estimator is

$$\hat{\theta}_T \in \arg \min_{\theta} \frac{1}{T} \sum_{t=0}^{T-1} \|X_{t+1} - f(X_t; \theta)\|^2. \quad (4.8)$$

We call

$$V_T(\theta) := \frac{1}{T} \sum_{t=0}^{T-1} \|X_{t+1} - f(X_t; \theta)\|^2 \quad (4.9)$$

the **prediction error criterion (PEC)**.

4.3 Linear Models: Consistency and Rates

Let

$$Z_t = \begin{cases} X_t & \text{for model (a),} \\ \Phi(X_t) & \text{for model (b),} \end{cases} \quad (4.10)$$

so that in both cases the model is linear in the unknown parameter: $X_{t+1} = \theta^* Z_t + W_t$.

Setting the gradient of V_T (with $f(X_t; \theta) = \theta Z_t$) to zero gives the normal equations $\sum_t (X_{t+1} - \hat{\theta}_T Z_t) Z_t^\top = 0$, i.e.,

$$\hat{\theta}_T = \left(\sum_{t=0}^{T-1} X_{t+1} Z_t^\top \right) \left(\sum_{t=0}^{T-1} Z_t Z_t^\top \right)^{-1}, \quad (4.11)$$

assuming the Gram matrix $\sum_t Z_t Z_t^\top$ is invertible. Noting that $X_{t+1} = \theta^* Z_t + W_t$,

$$\hat{\theta}_T - \theta^* = \left(\sum_{t=0}^{T-1} W_t Z_t^\top \right) \left(\sum_{t=0}^{T-1} Z_t Z_t^\top \right)^{-1} = \left(\frac{1}{T} \sum_{t=0}^{T-1} W_t Z_t^\top \right) \left(\frac{1}{T} \sum_{t=0}^{T-1} Z_t Z_t^\top \right)^{-1}. \quad (4.12)$$

💡 *Key Result 4.1*

Remark 4.7. The numerator converges to 0 almost surely, by the strong law of large numbers for martingale difference sequences [LW82]: each summand $W_t Z_t^\top$ satisfies $\mathbb{E}[W_t Z_t^\top | \mathcal{F}_t] = Z_t \mathbb{E}[W_t | \mathcal{F}_t]^\top = 0$ (Z_t is $\mathcal{F}_t$ -measurable, so it pulls out of the conditional expectation), so $\{W_t Z_t^\top\}_t$ is itself a matrix-valued martingale difference sequence.

We further assume that the denominator converges to a positive definite matrix $\Gamma \succ 0$:

Definition 4.8 (Persistence of excitation). The regressors $\{Z_t\}$ satisfy **persistence of excitation (PE)** if

$$\frac{1}{T} \sum_{t=0}^{T-1} Z_t Z_t^\top \longrightarrow \Gamma \succ 0 \quad \text{a.s.} \quad (4.13)$$

This is what is typically known as **persistence of excitation (PE)** in the literature.

Example 4.9 (LTI systems). Suppose $X_+ = AX + W$ with $W_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \Sigma)$, $\Sigma \succ 0$, and $\max_i |\lambda_i(A)| < 1$; suppose further that $X_0 = 0$. What does $\frac{1}{T} \sum_{t=0}^{T-1} X_t X_t^\top$ converge to as $T \rightarrow \infty$?

The stochastic process $\{X_t\}$ is **ergodic**: it has a unique stationary (invariant) distribution μ , and for any g with $\mathbb{E}_{X \sim \mu} |g(X)| < \infty$,

$$\frac{1}{T} \sum_{t=0}^{T-1} g(X_t) \longrightarrow \mathbb{E}_{X \sim \mu} [g(X)] \quad \text{a.s.}, \quad (4.14)$$

regardless of the initial condition X_0 . Therefore, if $\lim_{t \rightarrow \infty} \mathbb{E}[X_t X_t^\top] =: \Gamma$ exists, we have that $\frac{1}{T} \sum_{t=0}^{T-1} X_t X_t^\top \rightarrow \Gamma$ a.s.

Since $X_0 = 0$, $\mathbb{E}[X_1 X_1^\top] = \mathbb{E}[W_0 W_0^\top] = \Sigma =: \Gamma_1$, and, writing $\Gamma_t := \mathbb{E}[X_t X_t^\top]$,

$$\Gamma_{t+1} = \mathbb{E}[X_{t+1} X_{t+1}^\top] = \mathbb{E}[(AX_t + W_t)(AX_t + W_t)^\top] = A \mathbb{E}[X_t X_t^\top] A^\top + \mathbb{E}[W_t W_t^\top] = A \Gamma_t A^\top + \Sigma, \quad (4.15)$$

so that $\Gamma_+ = A \Gamma A^\top + \Sigma \implies \Gamma = A \Gamma A^\top + \Sigma$, since $\max_i |\lambda_i(A)| < 1$. Equivalently, in vectorized form,

$$\text{vec}(\Gamma_+) = (A \otimes A) \text{vec}(\Gamma) + \text{vec}(\Sigma), \quad (4.16)$$

a stable linear system driven by a constant input, converging to

$$\text{vec} \left(\sum_{k=0}^{\infty} A^k \Sigma (A^k)^\top \right) = \sum_{k=0}^{\infty} (A \otimes A)^k \text{vec}(\Sigma). \quad (4.17)$$

so that

$$\Gamma = \sum_{k=0}^{\infty} A^k \Sigma (A^k)^\top \quad (4.18)$$

is the **controllability Gramian** of $X_+ = AX + \Sigma^{1/2}V$: it captures how much energy from V , with V_t i.i.d. satisfying $\mathbb{E}V_t = 0$ and $\mathbb{E}V_t V_t^\top = I$, ends up distributed across each direction of the state space, and is a quantitative measure of **excitability**. (This is the same family of object as the **observability Gramian** from the *Dynamics, Stability, and Basic Lyapunov Theory* topic – there, $P = \sum_k (A^\top)^k Q A^k$ solved $A^\top P A - P + Q = 0$; here, Γ solves the dual equation $A \Gamma A^\top - \Gamma + \Sigma = 0$.)

Theorem 4.10 (Consistency, linear case). *Under persistence of excitation (4.13), $\hat{\theta}_T \rightarrow \theta^*$ almost surely.*

Proof. In the error decomposition (4.12), the numerator converges to 0 a.s. (martingale-difference SLLN) while the denominator converges to the positive definite matrix Γ a.s. (persistence of excitation, as verified for the LTI case in Example 4.9). By Slutsky's theorem, $\hat{\theta}_T - \theta^* \rightarrow 0 \cdot \Gamma^{-1} = 0$ a.s. $\square$

Exercise 4.11 (Linear Imitation Learning). Consider the **linear imitation learning** problem wherein $U_t^* = K_* X_t + N_t$, and K_* is such that the system $X_+^* = (A + BK_*)X^*$ is stable. Given a trajectory $\{X_t^*, U_t^*\}_t$ generated by the system $X_+^* = (A + BK_*)X^* + BN$, (i.e., $U_t^* = K_* X_t^* + N_t$), with noise $\{N_t\}$ satisfying Assumption 4.2, under what conditions on $A + BK_*$ and B is the PE condition eq. (4.13) satisfied? What is the limiting Gram matrix Γ in this case?

Everything else in this chapter is a variation on this decomposition: a mean-zero **numerator** divided by a growing **denominator**.

4.3.1 Asymptotic Rates of Convergence

We note that **asymptotic normality** allows us to provide an asymptotic convergence rate as well. In particular, instead of applying the martingale-difference SLLN to the numerator of (4.12), we instead use a central limit theorem for martingale difference sequences.

Remark 4.12 (Martingale CLT, specialized to our setting). A martingale difference sequence $\{\xi_t\}$ relative to $\mathcal{F}_t$ obeys a central limit theorem [LW82] whenever its *averaged conditional covariance* converges to a fixed limit,

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[\xi_t \xi_t^\top \mid \mathcal{F}_t] \longrightarrow V, \quad (4.19)$$

together with a mild regularity condition ruling out any single summand from dominating the sum (we won't spell this out here; see Lai and Wei [LW82]). The conclusion is $\frac{1}{\sqrt{T}} \sum_{t=0}^{T-1} \xi_t \xrightarrow{d} 0$

$\mathcal{N}(0, V)$ — this is the martingale central limit theorem in its general (quadratic-variation) form, but our martingale difference sequence has a particularly simple conditional covariance structure, which we can compute directly rather than invoking that machinery.

Take $\xi_t := \text{vec}(W_t Z_t^\top) = Z_t \otimes W_t$ (an MDS relative to $\mathcal{F}_t$, as noted above). Since Z_t is $\mathcal{F}_t$ -measurable, it pulls out of the conditional expectation, leaving only the noise covariance Σ from Assumption 4.2 behind:

$$\mathbb{E}[\xi_t \xi_t^\top | \mathcal{F}_t] = \mathbb{E}[(Z_t Z_t^\top) \otimes (W_t W_t^\top) | \mathcal{F}_t] = (Z_t Z_t^\top) \otimes \mathbb{E}[W_t W_t^\top | \mathcal{F}_t] = (Z_t Z_t^\top) \otimes \Sigma. \quad (4.20)$$

Averaging over t and invoking persistence of excitation (4.13),

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[\xi_t \xi_t^\top | \mathcal{F}_t] = \left(\frac{1}{T} \sum_{t=0}^{T-1} Z_t Z_t^\top \right) \otimes \Sigma \longrightarrow \Gamma \otimes \Sigma \quad \text{a.s.}, \quad (4.21)$$

so the martingale CLT above applies with $V = \Gamma \otimes \Sigma$:

$$\text{vec} \left(\frac{1}{\sqrt{T}} \sum_{t=0}^{T-1} W_t Z_t^\top \right) \xrightarrow{d} \mathcal{N}(0, \Gamma \otimes \Sigma), \quad \text{i.e.,} \quad \mathcal{Z} \stackrel{d}{=} \Sigma^{1/2} G \Gamma^{1/2}, \quad G_{ij} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1). \quad (4.22)$$

$$\sqrt{T}(\hat{\theta}_T - \theta^*) = \left(\frac{1}{\sqrt{T}} \sum_{t=0}^{T-1} W_t Z_t^\top \right) \left(\frac{1}{T} \sum_{t=0}^{T-1} Z_t Z_t^\top \right)^{-1} \xrightarrow{\text{Slutsky}} \mathcal{Z} \Gamma^{-1}, \quad (4.23)$$

where $\mathcal{Z} \Gamma^{-1} = \Sigma^{1/2} G \Gamma^{1/2} \Gamma^{-1} = \Sigma^{1/2} G \Gamma^{-1/2}$, so that

$$\text{vec}(\sqrt{T}(\hat{\theta}_T - \theta^*)) \xrightarrow{d} \mathcal{N}(0, \Gamma^{-1} \otimes \Sigma). \quad (4.24)$$

💡 Key Result 4.2

Remark 4.13 (An SNR-like quantity). We can extract a signal-to-noise-ratio-like quantity from this analysis. Note that Γ is roughly a measure of **signal energy**, while Σ is a measure of **noise energy**, so that

$$\|\Gamma^{-1} \otimes \Sigma\| = \|\Gamma^{-1}\| \cdot \|\Sigma\| = \frac{\lambda_{\max}(\Sigma)}{\lambda_{\min}(\Gamma)} =: \frac{1}{\text{SNR}}. \quad (4.25)$$

Example 4.14 (LTI system with isotropic noise). For $X_+ = AX + W$ with $W \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2 I)$ and $\Gamma = A \Gamma A^\top + \sigma^2 I$,

$$\text{SNR} = \frac{\lambda_{\min}(\Gamma)}{\sigma^2}, \quad (4.26)$$

where $\lambda_{\min}(\Gamma)$ is the least excitable direction in state space.

Remark 4.15. While this classical analysis captures important aspects of the statistical properties of the least-squares estimator, its guarantees only hold asymptotically, and hence capture neither dimensional dependence nor finite-data guarantees. Non-asymptotic bounds instead rely on concentration inequalities and are often more refined, giving convergence guarantees applicable at finite T . This has historically been the subject of much of this class: while very interesting from a theoretical perspective, well developed class notes, survey papers, and tutorial papers exist for the interested reader to learn from. See section 4.5 for some references on these contemporary results.

4.3.2 Sidebar: Stability Changes the Asymptotics

Consider the simplest possible instance of case (a): a scalar autoregression

$$X_{t+1} = aX_t + W_t, \quad W_t \stackrel{\text{iid}}{\sim} (0, \sigma^2). \quad (4.27)$$

Specializing (4.12) to this scalar case ($Z_t = X_t$),

$$\hat{a} - a = \frac{\sum_{t=0}^{T-1} X_t W_t}{\sum_{t=0}^{T-1} X_t^2}. \quad (4.28)$$

Definition 4.16 (O_p notation). $Z_T = O_p(r_T)$ means Z_T/r_T is bounded away from 0 and ∞ in probability as $T \rightarrow \infty$ — i.e., r_T captures the right **scale** of Z_T .

This is exactly the language needed to describe how fast $\sum_t X_t^2$ (the **signal**, in the language of section 4.3) grows in each regime:

Dynamics	Signal $\sum_t X_t^2$	Rate of $\hat{a} - a$	Limiting distribution
$ a < 1$ (stable)	$O_p(T)$	$T^{-1/2}$	Gaussian
$a = 1$ (unit root / marginal)	$O_p(T^2)$	T^{-1}	non-Gaussian
$ a > 1$ (explosive)	$O_p(a ^{2T})$	$ a ^{-T}$	non-Gaussian (Cauchy-type)

Stable case. When $|a| < 1$, X_t is (asymptotically) a stationary process with $\text{Var}(X_t) \rightarrow \sigma^2/(1 - a^2)$, so $\frac{1}{T} \sum_t X_t^2 \rightarrow \sigma^2/(1 - a^2)$ a.s. — this is exactly the persistence-of-excitation condition (4.13), holding in the scalar case with $\Gamma = \sigma^2/(1 - a^2)$. Plugging into Key Result 4.2 (with $\Sigma = \sigma^2$) gives

$$|a| < 1 : \quad \sqrt{T}(\hat{a} - a) \xrightarrow{d} \mathcal{N}(0, 1 - a^2). \quad (4.29)$$

💡 Key Result 4.3

Everything from section 4.3 applies unmodified.

Unit-root case (sketch). At $a = 1$, X_t is a random walk, so $X_t \sim \sqrt{t} \cdot (\text{something})$ and $\sum_t X_t^2$ grows like T^2 , not T — the information is accumulating **faster** than the general theory assumed, which is why the rate improves from $T^{-1/2}$ to T^{-1} :

$$a = 1 : \quad T(\hat{a} - 1) \xrightarrow{d} \frac{\int_0^1 W dW}{\int_0^1 W^2 dt}, \quad (4.30)$$

a ratio of functionals of Brownian motion W called the **Dickey-Fuller distribution**.

Explosive case (sketch). When $|a| > 1$, $X_t \approx a^t X_0$ eventually dominates any bounded contribution from the noise, so $\sum_t X_t^2 \sim O_p(a^{2T})$ — the information grows **geometrically** in T , which is why the rate improves all the way to $|a|^{-T}$:

$$|a| > 1 : \quad \hat{a} - a = O_p(|a|^{-T}). \quad (4.31)$$

With Gaussian innovations and $X_0 = 0$, an appropriately normalized explosive estimator has a Cauchy limiting distribution — heavier-tailed than **any** Gaussian, despite converging faster.

💡 Key Result 4.4

The same system $X_+ = aX + W$ with the same least-squares estimator can have radically different statistics based on the **stability of the underlying system**. This highlights something unique to learning for dynamical systems: the parameter being estimated often directly affects the statistics of the data!

Remark 4.17. Analogous statements can be made for the vector valued LTI system $X_+ = AX + W$. However, one must be careful in the **mixed stability setting**, in which some modes of A are stable, and some are unstable: in some cases, the OLS estimator is not statistically consistent! See [SR19] for more details.

4.4 Consistency Beyond Linear Parameterizations

Let's now move beyond linear models and consider the general nonlinear model

$$X_{t+1} = f(X_t; \theta^*) + W_t, \quad (4.32)$$

with the prediction error LS estimator $\hat{\theta}_T \in \arg \min_{\theta} \frac{1}{T} \sum_t \|X_{t+1} - f(X_t; \theta)\|^2$, and $V_T(\theta) := \frac{1}{T} \sum_t \|X_{t+1} - f(X_t; \theta)\|^2$ as before. The first thing we need to do is introduce a notion of well-posedness.

Definition 4.18 (Identifiability). θ^* is **identifiable** over a set Θ if it is the **unique** minimizer of the population criterion $V(\theta) := \lim_{T \rightarrow \infty} V_T(\theta)$ (limit a.s.) over Θ .

By ergodicity,

$$V(\theta) = \mathbb{E} \|X_+ - f(X; \theta)\|^2 = \mathbb{E} \|f(X; \theta^*) - f(X; \theta) + W\|^2 = \mathbb{E} \|f(X; \theta) - f(X; \theta^*)\|^2 + \mathbb{E} \|W\|^2, \quad (4.33)$$

where the cross term vanishes since W has conditional mean zero given $\mathcal{F}$ while $f(X; \theta) - f(X; \theta^*)$ is $\mathcal{F}$ -measurable. In particular, since $\mathbb{E} \|W\|^2$ doesn't depend on θ , identifying θ^* requires being able to **tell θ^* apart from θ using the data**: identifiability fails exactly when two different parameters produce the identical function $f(\cdot; \theta)$.

Example 4.19 (A non-identifiable model). Take $X_+ = (\theta_1 + \theta_2)X + W$. Every (θ_1, θ_2) with the same sum $\theta_1 + \theta_2$ produces the identical function $f(\cdot; \theta)$, so $V(\theta)$ only ever sees $\theta_1 + \theta_2$, and θ^* is not unique no matter how much data is collected.

Theorem 4.20 (Classical consistency, informal). *Suppose $V_T(\theta) \rightarrow V(\theta)$ almost surely and uniformly in θ , and θ^* is identifiable. Then $\hat{\theta}_T \rightarrow \theta^*$ almost surely.*

This is a version of the classical M-estimator consistency result based on a Wald-type argument [Jen69; Lai94].

4.4.1 Asymptotic Normality

Can we identify similar asymptotic rates and “SNR”-like quantities for this more general model? Our first task will be to define a notion of persistence of excitation for this more general model. To do so, we begin with a quantity which measures how responsive the function f is to changes in the parameter θ near the true parameter θ^* .

Definition 4.21 (Sensitivity). The **sensitivity** of a model $f(X; \theta)$ at state X_t around θ^* is

$$J_t := \left. \frac{\partial f(X_t; \theta)}{\partial \theta} \right|_{\theta=\theta^*}. \quad (4.34)$$

Remark 4.22 (Connection to the score). The sensitivity J_t is closely related to the likelihood score. For example, under the Gaussian model $X_{t+1} | X_t \sim \mathcal{N}(f(X_t; \theta), \Sigma)$, the one-step score is $s_t(\theta) := \nabla_{\theta} \log p_{\theta}(X_{t+1} | X_t) = \left(\frac{\partial f(X_t; \theta)}{\partial \theta} \right)^{\top} \Sigma^{-1} (X_{t+1} - f(X_t; \theta))$. Evaluating at the true parameter θ^* gives $s_t(\theta^*) = J_t^{\top} \Sigma^{-1} W_t$, since $J_t = \left. \frac{\partial f(X_t; \theta)}{\partial \theta} \right|_{\theta=\theta^*}$ and $X_{t+1} - f(X_t; \theta^*) = W_t$. Thus J_t maps the one-step prediction error—which at the true parameter is precisely the noise W_t —into a parameter-space score direction. In particular, $\mathbb{E}[s_t(\theta^*) s_t(\theta^*)^{\top} | X_t] = J_t^{\top} \Sigma^{-1} J_t$, so greater sensitivity of the dynamics to θ corresponds to greater local information about the parameter.

We assume the following persistence-of-excitation-like condition on the sensitivity $\{J_t\}$:

Assumption 4.23 (Persistence of Excitation (Nonlinear)). The sensitivity $\{J_t\}$ satisfies

$$H := \mathbb{E}_{X \sim \pi} [J_t^{\top} J_t] \succ 0. \quad (4.35)$$

We call H the **Gauss–Newton information matrix** [Jen69; Lai94].

Example 4.24. For a model that is linear in the parameters, $f(Z_t; \theta) = \theta Z_t$, $J_t = \partial f(Z_t; \theta) / \partial \theta|_{\theta=\theta^*} = Z_t$ and $H = \mathbb{E}[Z_t Z_t^{\top}] = \Gamma \succ 0$ from section 4.3. Everything below is the nonlinear extension of this “signal” matrix.

Let’s see why this is a useful notion of excitation for the general model $f(X; \theta)$. In what follows, we assume θ is a vector in $\mathbb{R}^p$.

Theorem 4.25 (Asymptotic normality). *Under Assumption 4.23 and suitable ergodicity assumptions,*

$$\sqrt{T}(\hat{\theta}_T - \theta^*) \xrightarrow{d} \mathcal{N}(0, H^{-1} \Omega H^{-1}), \quad \Omega := \mathbb{E}_{X \sim \pi} [J_t^{\top} \Sigma J_t]. \quad (4.36)$$

In the case of homoskedastic noise with $\Sigma = \sigma^2 I$, we have that $\Omega = \sigma^2 H$ and thus this collapses to $\mathcal{N}(0, \sigma^2 H^{-1})$.

Proof. The estimator $\hat{\theta}_T$ satisfies the first-order condition $\nabla V_T(\hat{\theta}_T) = 0$. Taylor-expanding around θ^* ,

$$0 = \nabla V_T(\hat{\theta}_T) \approx \nabla V_T(\theta^*) + \nabla^2 V_T(\theta^*)(\hat{\theta}_T - \theta^*), \quad (4.37)$$

so that $\hat{\theta}_T - \theta^* \approx -[\nabla^2 V_T(\theta^*)]^{-1} \nabla V_T(\theta^*)$.

Recall $V_T(\theta) = \frac{1}{T} \sum_t \|X_{t+1} - f(X_t; \theta)\|^2$, so

$$\nabla V_T(\theta) = -\frac{2}{T} \sum_{t=0}^{T-1} \frac{\partial f(X_t; \theta)^\top}{\partial \theta} (X_{t+1} - f(X_t; \theta)) \implies \nabla V_T(\theta^*) = -\frac{2}{T} \sum_{t=0}^{T-1} J_t^\top W_t, \quad (4.38)$$

$$\nabla^2 V_T(\theta) = \frac{2}{T} \sum_{t=0}^{T-1} \frac{\partial f^\top}{\partial \theta} \frac{\partial f}{\partial \theta} - \frac{2}{T} \sum_{t=0}^{T-1} \frac{\partial^2 f(X_t; \theta)}{\partial \theta^2} \cdot (X_{t+1} - f(X_t; \theta)) \quad (4.39)$$

where the second identity uses $X_{t+1} - f(X_t; \theta) = W_t$ at $\theta = \theta^*$, so

$$\nabla^2 V_T(\theta^*) = \frac{2}{T} \sum_{t=0}^{T-1} J_t^\top J_t - \frac{2}{T} \sum_{t=0}^{T-1} \frac{\partial^2 f}{\partial \theta^2} \Big|_{\theta=\theta^*} \cdot W_t \longrightarrow 2H, \quad (4.40)$$

since $\frac{2}{T} \sum_t J_t^\top J_t \rightarrow 2H$ by the same ergodic-averaging argument used for Γ above, and the second term vanishes (it is again a martingale-difference average, exactly as in section 4.3). So now we have

$$\sqrt{T}(\hat{\theta}_T - \theta^*) \approx \left[\frac{1}{T} \sum_{t=0}^{T-1} J_t^\top J_t \right]^{-1} \frac{1}{\sqrt{T}} \sum_{t=0}^{T-1} J_t^\top W_t. \quad (4.41)$$

By suitable ergodicity assumptions, $\left[\frac{1}{T} \sum_t J_t^\top J_t \right]^{-1} \rightarrow H^{-1}$, and by the martingale-difference CLT, $\frac{1}{\sqrt{T}} \sum_t J_t^\top W_t \xrightarrow{d} \mathcal{N}(0, \Omega)$ with $\Omega := \mathbb{E}[J_t^\top \Sigma J_t]$. Combining via Slutsky's theorem gives $\sqrt{T}(\hat{\theta}_T - \theta^*) \xrightarrow{d} H^{-1} \mathcal{N}(0, \Omega) = \mathcal{N}(0, H^{-1} \Omega H^{-1})$. $\square$

Remark 4.26 (An SNR-like bound). We can identify a similar SNR-like term here as well, by looking at

$$\frac{1}{\|H^{-1} \Omega H^{-1}\|} \geq \frac{\lambda_{\min}^2(H)}{\lambda_{\max}(\Omega)} \quad (4.42)$$

(here Ω is how the noise interacts with the sensitivity of the model). Unlike the linear case, this is only a **lower bound**, not an exact expression: H and Ω no longer combine into the clean Kronecker structure that made Key Result 4.2 exact. It becomes exact again in the homoskedastic case, $\Sigma = \sigma^2 I$: there $\Omega = \sigma^2 H$, so $H^{-1} \Omega H^{-1} = \sigma^2 H^{-1}$ exactly, and

$$\text{SNR} := \frac{1}{\|H^{-1} \Omega H^{-1}\|} = \frac{\lambda_{\min}(H)}{\sigma^2}. \quad (4.43)$$

Aside: Fisher Information

$H = \mathbb{E}[J_t^\top J_t]$ depends only on the model's sensitivity J_t ; it says nothing about the **shape** of the noise distribution, only implicitly about its covariance (via Ω). The **Fisher information**

$$\mathcal{I}(\theta^*) := \mathbb{E} \left[\nabla_\theta \log p_{\theta^*}(X_{t+1} | X_t) \nabla_\theta \log p_{\theta^*}(X_{t+1} | X_t)^\top \right], \quad (4.44)$$

by contrast, is built from the actual conditional likelihood p_θ — it depends on the full noise density, not just its second moment.

They coincide exactly when the noise is Gaussian and isotropic. If $W_t | \mathcal{F}_t \sim \mathcal{N}(0, \sigma^2 I)$, the conditional log-likelihood is $\log p_\theta(X_{t+1} | X_t) = -\frac{1}{2\sigma^2} \|X_{t+1} - f(X_t; \theta)\|^2 + \text{const}$ — i.e., **least**

squares is exactly the conditional MLE — and the score at θ^* is $\nabla_{\theta} \log p_{\theta^*} = \sigma^{-2} J_t^{\top} W_t$. A direct computation using $\mathbb{E}[W_t W_t^{\top} | \mathcal{F}_t] = \sigma^2 I$ gives

$$\mathcal{I}(\theta^*) = \mathbb{E}[\sigma^{-4} J_t^{\top} W_t W_t^{\top} J_t] = \sigma^{-2} \mathbb{E}[J_t^{\top} J_t] = \sigma^{-2} H, \quad (4.45)$$

so $\sigma^2 H^{-1} = \mathcal{I}(\theta^*)^{-1}$: the homoskedastic case of Theorem 4.25 is the Cramér–Rao-efficient covariance. This is why the homoskedastic special case looks exactly like a textbook Fisher-information formula: it secretly assumed isotropic Gaussian noise all along, since that’s the only case in which minimizing squared error and maximizing likelihood are the same optimization problem.

And if the noise is homoskedastic but not Gaussian? Nothing above breaks — Theorem 4.20 never needed Gaussianity, and Theorem 4.25 only used the second moment of W_t — so $\hat{\theta}_T$ is still consistent and asymptotically normal with covariance $\sigma^2 H^{-1}$. But that covariance is generally **not** $\mathcal{I}(\theta^*)^{-1}$ anymore: by the Cramér–Rao bound, $\sigma^2 H^{-1} \succeq \mathcal{I}(\theta^*)^{-1}$, with equality iff the noise is Gaussian. Least squares is consistent for **any** zero-mean, finite-variance noise, but it is only statistically **efficient** when the noise happens to be Gaussian — otherwise it is implicitly optimizing the wrong (Gaussian) log-likelihood, leaving information on the table. To attain the Cramér–Rao bound, one can instead solve the **maximum likelihood estimation problem** induced by the noise process $\{W_t\}_t$.

4.5 Further Reading

The classical reference for the linear/PE story of section 4.3 is Lai and Wei [LW82], which proves strong consistency and asymptotic normality for least squares with stochastic regressors, applied explicitly to identification and control of dynamic systems. The nonlinear extension in section 4.4 draws on Jennrich [Jen69] (classical nonlinear-least-squares consistency/asymptotic-normality theory) and Lai [Lai94], which extends nonlinear least squares to stochastic regressors with martingale-difference disturbances, covering case (c) directly. The scalar stable/unit-root/explosive sidebar of section 4.3.2 is the classical result of Anderson [And59], who treats all three regimes in one paper, including the Gaussian $\sqrt{T}$ result in the stable regime, non-normal behavior at $|a| = 1$, and Cauchy-type explosive limits; Lai and Wei [LW83] later proved strong consistency of least squares for general autoregressive models **without** restricting the characteristic roots to be stable — i.e., consistency survives across all three rows of the table even though the rate and limit law do not.

The $\sqrt{T}$ rate of Key Result 4.2 is purely **asymptotic**: it is a statement about what happens as $T \rightarrow \infty$, and says nothing about how large T must be before the Gaussian limit is a good approximation. A more modern line of work replaces the martingale central limit theorem with a **self-normalized martingale concentration inequality** [APS11] (originally developed for linear bandits), and obtains high probability error bounds on $\|\hat{\theta}_T - \theta^*\|$ which scale as $O\left(\sqrt{\frac{d \log T + \log(1/\delta)}{T}}\right)$, where $d := \dim \theta$. Again, these results only hold under suitable stability assumptions on the underlying data generating process. For example, Simchowitz et al. [Sim+18] builds on these tools to obtain sharp finite-sample rates for linear system identification.

For a tutorial-length treatment of this and related non-asymptotic techniques — mixing-free analysis of unstable systems, learning from many independent trajectories, and more — see Ziemann et al. [Zie+23], and, for a broader survey of the finite-sample perspective on identification and control, Tsiamis et al. [Tsi+23]. Ingvar Ziemann’s lecture notes for a past

offering of this course Ziemann [Zie24] develop this non-asymptotic theory in depth, and are the natural next stop for a reader who wants to go beyond the classical, asymptotic story told in this chapter.

References

- [APS11] Yasin Abbasi-Yadkori, Dávid Pál, and Csaba Szepesvári. “Improved Algorithms for Linear Stochastic Bandits”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2011, pp. 2312–2320.
- [And59] T. W. Anderson. “On Asymptotic Distributions of Estimates of Parameters of Stochastic Difference Equations”. In: *Annals of Mathematical Statistics* 30.3 (1959), pp. 676–687.
- [Jen69] R. I. Jennrich. “Asymptotic Properties of Non-Linear Least Squares Estimators”. In: *Annals of Mathematical Statistics* 40.2 (1969), pp. 633–643.
- [Lai94] T. L. Lai. “Asymptotic Properties of Nonlinear Least Squares Estimates in Stochastic Regression Models”. In: *Annals of Statistics* 22.4 (1994), pp. 1917–1930.
- [LW82] T. L. Lai and C. Z. Wei. “Least Squares Estimates in Stochastic Regression Models with Applications to Identification and Control of Dynamic Systems”. In: *Annals of Statistics* 10.1 (1982), pp. 154–166.
- [LW83] T. L. Lai and C. Z. Wei. “Asymptotic Properties of General Autoregressive Models and Strong Consistency of Least-Squares Estimates of Their Parameters”. In: *Journal of Multivariate Analysis* 13.1 (1983), pp. 1–23.
- [SR19] Tuhin Sarkar and Alexander Rakhlin. “Near optimal finite time identification of arbitrary linear dynamical systems”. In: *International Conference on Machine Learning*. PMLR. 2019, pp. 5610–5618.
- [Sim+18] Max Simchowitz, Horia Mania, Stephen Tu, Michael I. Jordan, and Benjamin Recht. “Learning Without Mixing: Towards a Sharp Analysis of Linear System Identification”. In: *Proceedings of the 31st Conference on Learning Theory (COLT)*. Vol. 75. PMLR. 2018, pp. 439–473.
- [Tsi+23] Anastasios Tsiamis, Ingvar Ziemann, Nikolai Matni, and George J. Pappas. “Statistical Learning Theory for Control: A Finite-Sample Perspective”. In: *IEEE Control Systems Magazine* 43.6 (2023), pp. 67–97. DOI: [10.1109/MCS.2023.3310345](https://doi.org/10.1109/MCS.2023.3310345).
- [Zie24] Ingvar Ziemann. *Lecture Notes for ESE6180: Learning, Dynamics and Control*. Oct. 2024. URL: https://ingvarziemann.com/ESE6180/Lecture_Notes_ESE_6180.pdf (visited on 08/21/2026).
- [Zie+23] Ingvar M. Ziemann, Anastasios Tsiamis, Bruce D. Lee, Yassir Jedra, Nikolai Matni, and George J. Pappas. “A Tutorial on the Non-Asymptotic Theory of System Identification”. In: *62nd IEEE Conference on Decision and Control (CDC)*. Singapore, 2023, pp. 8921–8939.